

Channel-Aware Cluster-Aware Multivariate Time Series Classification Framework on Reduced Representation

1 Exploratory Data Analysis (EDA) Plan

The exploratory data analysis (EDA) phase aims to characterize multivariate time series (MTS) datasets through a unified and reproducible methodology. Instead of analyzing all datasets individually, a representative dataset is selected from each of five major categories to reflect the diversity of sensing modalities and temporal behaviors found in multivariate time series. This design ensures that the same set of analytical steps can generalize across domains while minimizing redundancy.

1.1 Dataset Categories

To capture the full range of temporal and spatial variability present in the UEA 30 benchmark, datasets are grouped into five main categories according to their sensing modality and task characteristics:

1. **Wearable IMU and Human Activity:** Datasets collected from inertial or motion sensors representing body movements and gestures. Examples include *Basic-Motions*, *Cricket*, *Libras*, and *NATOPS*. A single representative dataset, **NATOPS**, is used for detailed EDA.
2. **Brain Signal (EEG/MEG):** Neural recordings capturing brain activity through electroencephalography or magnetoencephalography. Typical datasets include *MotorImagery*, *FingerMovements*, and *SelfRegulationSCP1/2*. The representative dataset chosen is **MotorImagery**.
3. **Audio and Spectral:** Time–frequency representations of acoustic signals such as speech, heartbeats, or animal sounds. Representative datasets include *InsectWingbeat*, *SpokenArabicDigits*, and *JapaneseVowels*, with **InsectWingbeat** selected for analysis.
4. **Trajectory and Handwriting:** Spatial sequences describing motion paths or pen traces, such as *CharacterTrajectories* and *PenDigits*. The representative dataset for this category is **CharacterTrajectories**.
5. **Environmental and Physical Systems:** High-dimensional sensor data describing physical or environmental processes such as traffic, energy consumption, or chemical concentration. Datasets include *PEMS-SF*, *EthanolConcentration*, and *LSST*, with **PEMS-SF** selected for EDA.

Each representative dataset undergoes the same statistical and visual analyses described in the subsequent sections, enabling consistent comparison and category-level generalization across the full benchmark.

1.2 EDA Pipeline

The following analyses are uniformly applied to every dataset within the framework, independent of its domain or sensing modality. Each stage extracts reproducible statistics that collectively describe the structural, temporal, and relational characteristics of the data.

1. Dataset Overview Provides a summary of the dataset structure, including the number of samples (N), timesteps (T), channels (C), classes (K), and datatype. This step establishes a foundational understanding of dataset size, dimensionality, and class composition.

2. Sequence Length Distribution Examines the distribution and descriptive statistics of sequence lengths, such as mean, median, minimum, maximum, and standard deviation. This analysis determines whether the dataset contains fixed-length or variable-length time series and highlights any inconsistencies requiring preprocessing or normalization.

3. Channel-Level Descriptive Statistics Computes core statistics for each channel, including the mean, standard deviation, skewness, and kurtosis. These metrics describe the variability and distributional characteristics of each sensor, helping to identify inactive, noisy, or unstable channels.

4. Channel Correlation Structure Evaluates pairwise correlations between channels, aggregated across all samples and timesteps. This step identifies inter-channel dependencies, redundancy among sensors, and potential structural relationships such as symmetry or coupling effects.

5. Class-Wise Temporal Profiles Calculates the average and dispersion of temporal signals for each class label. This highlights how classes differ in their temporal dynamics and whether their trajectories exhibit distinct statistical or amplitude patterns over time.

6. Feature-Space Projection (Separability) Derives global statistical features for each sample (e.g., channel-wise mean and standard deviation) and analyzes their distribution in a reduced feature space. This serves to estimate the degree of separability and potential clustering of classes under simple statistical transformations.

7. Autocorrelation Function (ACF) Measures the temporal dependency of each channel by computing its autocorrelation across fixed time lags. This reveals the degree of temporal persistence, smoothness, and periodicity present within the recorded signals.

8. Channel Variance Ranking Quantifies and ranks the overall variance of each channel across all samples and timesteps. Channels exhibiting high variability are often more informative and can guide channel selection or dimensionality reduction within the proposed classification framework.